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Graduation Project Dissertation

to obtain the National Diploma of Applied License in Development of Embedded and Mobile Systems

Dev and Deployment of an automated real-time monitoring system for TESCA group

Carried out by :
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Defended on 00/00/2026, in front of the committee members :

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End of studies internship completed at

Host company TS2I Technique Solution et Innovation Industriel

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Graduation Project in Development of Embedded and Mobile Systems	
Made by: Fedi Amdouni Firas Ouesleti	**Batch:** 2025-2026
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Summary:

In modern manufacturing, accurate production monitoring is essential for optimizing efficiency. However, many facilities face unreliable traceability of machine stoppages and their root causes, hindering precise Overall Equipment Effectiveness (OEE) calculation and performance analysis. This project addresses this challenge by designing and deploying an automated, real-time industrial monitoring system. Built on a multi-layer architecture. The system automatically detects stoppages, enables real-time cause logging, and centralizes production data in a MySQL database, while analytical dashboards provide dynamic KPI visualization. Implementation results demonstrate complete machine event traceability, automatic OEE calculation per shift and day, and a significant reduction in manual entry errors and undetected production losses. The system successfully enhances production transparency and decision-making.

Keywords: Industrial monitoring, OEE, Modbus TCP, PLC, Node-RED, NestJS, HMI, machine stoppages, MySQL database

Résumé:

Dans l'industrie moderne, une surveillance précise de la production est essentielle pour optimiser l'efficacité. Cependant, de nombreuses installations font face à une traçabilité peu fiable des arrêts machine et de leurs causes profondes, entravant le calcul précis de l'Efficacité Globale des Équipements (OEE) et l'analyse des performances. Ce projet relève ce défi en concevant et déployant un système de surveillance industrielle automatisé et en temps réel. Construite sur une architecture multi-couches, . Le système détecte automatiquement les arrêts, permet l'enregistrement en temps réel des causes, et centralise les données de production dans une base de données MySQL, tandis que des tableaux de bord analytiques fournissent une visualisation dynamique des indicateurs de performance (KPI). Les résultats de l'implémentation démontrent une traçabilité complète des événements machine, un calcul automatique de l'OEE par équipe et par jour, et une réduction significative des erreurs de saisie manuelle et des pertes de production non détectées. Le système améliore avec succès la transparence de la production et la prise de décision, avec des travaux futurs prévus pour l'extension à d'autres lignes de production.

Mots-clés: Surveillance industrielle, OEE, Modbus TCP, PLC, Node-RED, NestJS, IHM, arrêts machine, base de données MySQL

Dedication

First and foremost, I express my deepest gratitude to Allah for His infinite blessings, wisdom, and strength throughout this journey. His guidance has been my compass in overcoming challenges and achieving this milestone.

To my favorite cheerleaders in life,

To my parents,

The architects of my dreams and the guardians of my laughter. Your love has painted my world with vibrant hues of happiness and warmth. This piece is just a little way to show how much I appreciate all the sacrifices you've made and the endless love you've given me.

To my brother,

The partner in crime,with your playful banter and unwavering support, you've made every moment an adventure worth living. This dedication is a reminder of the special bond we share and the joy you bring into my life.

To my dear friends,

The stars that light up my sky on the darkest of nights. Your friendship is my greatest treasure. This work is dedicated to you as a symbol of our enduring bond and the countless memories we've yet to create.

May this dedication serve as a humble token of my gratitude to all those who have touched my life with their kindness, wisdom and love.

Fedi Amdouni

Dedication

First and foremost, I would like to thank God for all His blessings upon me and my family. For the strength, patience, and courage He granted me each day to pursue my goals and accomplish this final year project.

I dedicate this humble work to:

To my beloved parents,

Your unwavering support, endless sacrifices, and constant prayers have been the foundation of my success. May Allah grant you long life, good health, and the happiness you truly deserve.

To my dear sister,

For your encouragement, understanding, and the joy you bring to my life. Your belief in me has been a constant source of motivation.

To my partner, Fedi Amdouni,

For your dedication, teamwork, and perseverance throughout this project. This achievement is as much yours as it is mine.

To all my friends and mentors,

Who have guided, supported, and inspired me along the way. Your wisdom and kindness have shaped my path.

To everyone who contributed to this journey, directly or indirectly, I offer my sincere gratitude.

Firas Ouesleti

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Foremost, we extend our sincere appreciation to **Mr. Radhouane Aloui**, our university supervisor.His expert guidance, unwavering support and insightful mentorship have been instrumental throughout this endeavor. His constant feedback and encouragement pushed us to exceed our limits, particularly in exploring advanced Automation and embedded systems concepts, and we are truly grateful for his dedication to our academic and professional growth.

We are equally indebted to our industry supervisor, **Mr. Fethi Yahyaoui**, for his invaluable presence and hands-on guidance during our internship at TESCA Group. His willingness to explain complex industrial processes, challenge our critical thinking and bridge the gap between theoretical knowledge and real-world application has been profoundly inspiring. We deeply appreciate his commitment to fostering our development both as studrnts and as professionals.

Our gratitude also extends to the entire team at TESCA Group for providing us with a stimulating, collaborative and supportive work environment. The opportunity to apply our skills in a dynamic industrial setting has been an invaluable learning experience that will undoubtedly shape our future careers.

We would like to thank the faculty and staff of ISET Beja for the strong educational foundation they have provided us over the past three years. The knowledge, technical skills and professional values we have acquired within this institution have been the cornerstone of this project.

We sincerely thank the members of the jury for taking the time to evaluate this work. We deeply appreciate your efforts and look forward to your constructive feedback and insights.

Finally, we extend our heartfelt thanks to everyone who supported us directly or indirectly throughout this journey. Your encouragement, patience and belief in our potential have been our greatest motivation. This achievement is as much yours as it is ours.

Fedi Amdouni & Firas Ouesleti

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General Introduction

In today's competitive industrial landscape, production efficiency is no longer a luxury, it is a necessity. For manufacturing firms, production efficiency is essential because their production process faces numerous challenges, and they must be ready at all times to optimize production output, reduce wastage, and mitigate any disturbances that may arise within the production floor. One of the critical indicators of the effectiveness of production processes is the Overall Equipment Effectiveness (OEE), known in French as the *Taux de Rendement Synthétique* (TRS), stands out as one of the most comprehensive metrics, capturing the combined impact of availability, performance, and quality losses in the production process. [1].

The automotive industry, governed by rigorous quality standards such as IATF 16949 [2], is particularly demanding in this regard. Tunisia has established itself over the decades as a competitive industrial hub for automotive suppliers, attracting major international players thanks to its geographic proximity to Europe, its cost-efficient production and skilled workforce [3]. Among these companies is TESCA Group, an international manufacturer specializing in vehicle interior components, with several production sites operating across Tunisia [4].

TESCA Group has always prioritized continuous improvement. Better visibility into production matters—detecting stoppages quickly, understanding what causes them, and responding fast directly impact competitiveness. To achieve this, they developed a production monitoring application that automates data collection and OEE calculation.

But operational use revealed real problems with the Node-RED setup. The dashboard can't be customized, visualizations are limited, and it doesn't scale. Integration is restricted. The company also has no mechanical or electrical documentation, which means maintenance teams can't diagnose equipment problems or plan maintenance work properly. These gaps stop the system from supporting TESCA Group's improvement goals.

Our Final Year Project tackled these issues in three ways. We built a modern dashboard to replace the old interface—one that's easier to use and lets teams customize what they see. We rewrote the backend from scratch on a proper server, which means better performance and reliability. We also put together complete documentation for the mechanical and electrical systems so maintenance crews can actually fix things without guessing. The result is a monitoring system that gives TESCA Group real visibility into what's happening on the floor, helps them spot problems faster, and makes their continuous improvement work actually possible.

This report summarizes all the work completed during this Final Year Project.It documents five key areas of development: first, preliminary analysis presenting the two hosting organizations and covering the project context and the limitations of the existing Node-RED implementation. Second, the development of a modern, dedicated frontend dashboard with professional data visualization and customizable layouts. Third, the backend architecture migration and automation programming, and HMI logic development. Fourth, the database optimization to improve system performance, reliability, and scalability. And finally, the creation of comprehensive maintenance plans and technical documentation for mechanical and electrical systems to support long-term operations.

Chapter 1

Preliminary analysis

Introduction

The first chapter of this report aims to provide an overview of the host and the client companies, present the case study by highlighting the problem to be addressed, and introduce the modeling language used for the project. This chapter sets the stage for the subsequent sections by establishing the context, significance, and scope of the actions undertaken in the next chapters.

1 Presentation of the enterprises

1.1 TS2I

TS2I is a leading company in the field of specialised machines fabrication, automation, industrial maintenance, and Tampographique service. With a strong commitment to quality, innovation, and customer satisfaction, TS2I has established itself as a trusted partner for various industries. The following provides an overview of TS2I and highlights its expertise in fabrication machines, automation, industrial maintenance, and pad printing services.

- **Fabrication of specialised machines:** TS2I specializes in the design, development, and production of fabrication of specialised machines. These machines are tailored to meet the unique requirements of clients in diverse industries such as automotive, aerospace, electronics, and more. With a team of experienced engineers and technicians, TS2I leverages cutting-edge technologies to deliver custom fabrication machines that optimize productivity, efficiency, and quality. Whether it's automated assembly lines, robotic systems, or specialized machining equipment, TS2I ensures that each machine is built to the highest standards.

- **Automation:** Automation plays a crucial role in enhancing productivity and reducing operational costs. TS2I offers comprehensive automation solutions to help businesses automate their processes. From simple control systems to complex integrated automation solutions, TS2I designs and implements tailored automation solutions based on clients' requirements. Their expertise in PLC programming, HMI design, robotics, and motion control allows them to deliver efficient and reliable automation solutions that streamline operations and improve overall efficiency.

- **Industrial maintenance:** TS2I understands the importance of maintaining industrial equipment to ensure optimal performance and minimize downtime. The company offers comprehensive maintenance services to address the unique needs of its clients. Whether it's preventive maintenance, troubleshooting, or equipment repair, TS2I's skilled technicians and engineers are well-equipped to handle a wide range of industrial machinery. By providing regular maintenance, TS2I helps clients avoid costly breakdowns, increase equipment lifespan, and maximize productivity.

- **Tampographique service:** In addition to its fabrication machines and automation expertise, TS2I also offers a specialized service in tampography, or pad printing. Tampography is a versatile printing technique used for applying designs or logos onto various surfaces, including plastics, metals, ceramics, and glass. TS2I's pad printing service ensures high-quality and precise printing, providing clients with branding solutions that are durable and visually appealing. Whether it's product customization, promotional items, or industrial labeling, TS2I's pad printing service offers reliable and efficient solutions.

1.2 TESCA

1.2.1 Presentation of the company

Tesca Group is a family-run company with a background in textiles that has successfully ventured into the automotive industry, specifically automotive seating. Emulating the approach of a French automotive manufacturing house, Tesca Group meticulously crafts each piece, from its atelier to its global production sites[4]. The company strives to embody the same creative audacity and prides itself on being a partner to major automotive manufacturers like:

Figure 1: Tesca's partners

With a clientele consisting of demanding visionaries who shape the future of mobility, Tesca Group positions itself as a reliable and flexible partner, accompanying its clients in their pursuit of innovation and competitiveness. The company believes that adopting a long-term and eco-friendly vision is crucial for sustainable growth, constantly pushing for excellence rather than settling for mediocrity[4].

- **Sustainability:** A concrete and integral part of Tesca Group's approach. The company is committed to sustainable development and actively incorporates environmental and social values into its products, services, and activities within the automotive industry. Since 2004, Tesca Group has been dedicated to controlling its ecological footprint, and the executive management maintains and expands upon these efforts. The company adheres to the values and principles of the Global Compact through its Ethics Charter, to obtain ISO 14001 certification for all its sites[4].

- **Environmental policy:** Tesca Group's environmental policy revolves around five major directions: behaving in an environmentally responsible manner, optimizing and controlling the industrial waste, optimizing energy consumption and natural resources, participating in the reduction of the ecological footprint throughout the design and production processes, and ensuring compliance with environmental regulations and requirements[4].

- **Innovation:** A driving force within Tesca Group, with a constant emphasis on generating new ideas and executing them effectively. The company's research programs focus on key themes in the automotive industry, such as weight reduction, optimized part design, and ecological footprint expertise, while prioritizing user comfort, functionality, and safety. Tesca Group holds numerous patents in the conception, design, and manufacturing of automotive parts and seat components, including headrests, armrests, upholstery, padding, and "smart textiles"[4].

- **Industrial excellence:** A cornerstone of Tesca Group's strategy, whether through its local production sites or research centers. The company strives to provide reliable processes that meet the evolving market expectations worldwide. Tesca Group empowers its teams through a culture of continuous improvement, with competitiveness, efficiency, and exceptional service forming the foundation of its industrial success. SPRINT (Tesca's Industrial Production System) is employed to engage collaborators at all levels, aiming to standardize procedures and ensure perpetual progress in industrial performance. Shared values among Tesca Group employees include excellence, self-discipline, commitment, autonomy, and pragmatism[4].

- **Quality:** Tesca Group's Quality policy aligns with five major directions: meeting commitments to conformity, cost, and deadlines; aligning industrial and purchasing performance with the highest quality criteria; standardizing existing processes to offer innovative products; prioritizing process control and prevention; and continuously enhancing the skills of its teams[4].

- **History:** In terms of its history, Tesca Group has a notable timeline of achievements. Starting in 1960 with the provision of Citroën 2CV roofs, the company went on to pioneer and apply the Foam In Place technology in 1978. In 1983, Tesca Group established its Design Studio in Paris, followed by the creation of the CERA research and development center in Reims in 1993. By the year 2000, the company had expanded its production capabilities worldwide. In 2016, Tesca Group underwent a significant transformation, becoming Trèves TSC[4].

Overall, Tesca Group has established itself as a reputable partner to major automotive manufacturers, driven by a commitment to sustainability, innovation, industrial excellence, and quality. Through its diverse range of products and services, the company aims to shape the future[4].

1.2.2 Technical file

The following table presents the technical file of TESCA.

Table 1: Technical file of TESCA

Company name	**TESCA**
Sector	Vehicle interior equipment
Creation	2019
Legal form	Foreign company
Staff	179 employees
Address	Grombalia Industrial Zone – Nabeul– Tunisia

1.2.3 Organizational chart

The organizational chart shown in Figure 2 presents the distribution of positions and functions within the Tesca enterprise.

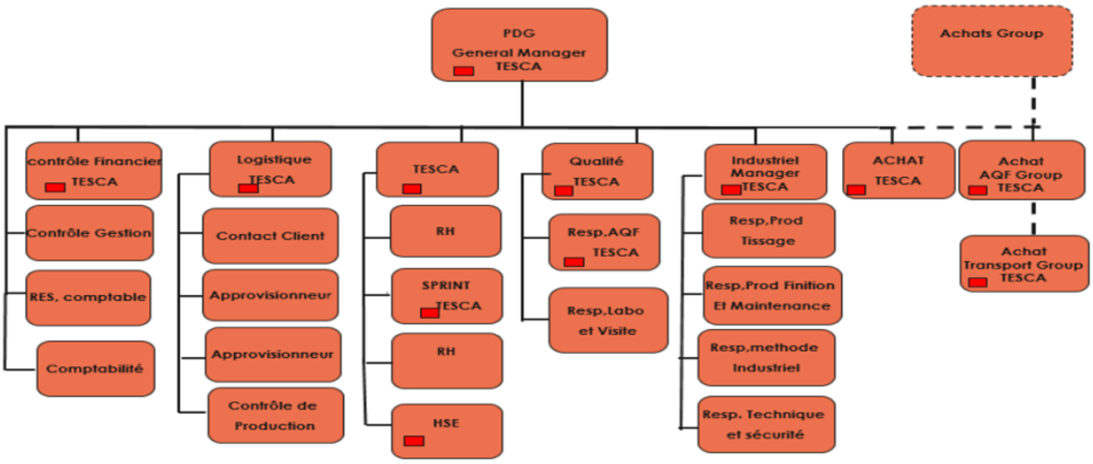

Figure 2: Organizational chart of TESCA

2 Case study

2.1 Description of the existing system

2.1.1 Overview of the Existing System

The existing production monitoring system was developed in the previous internship project to replace manual paper-based stop recording with automated digital tracking. The system consists of three components working together: a Delta DVP PLC installed on the fabric inspection machine, a Node-RED platform serving as the intermediary that reads PLC data and communicates with the database, and a MySQL database storing all production information.

Before this system existed, operators manually recorded all machine stops, causes, and durations on paper during their shifts. This manual approach made monitoring difficult, introduced frequent errors, and prevented real-time analysis. The new system automates data collection: when the machine stops, the PLC detects it, the operator selects the cause through a tablet interface, and the system automatically calculates the duration when the machine restarts.

The system serves two main user groups. Operators on the production floor use a tablet application that displays real-time machine speed, yardage measurements, lighting controls, and a dropdown menu to select stop causes. This interface updates continuously and provides immediate feedback, making it practical for shop-floor use. Managers access a web interface where they can filter data by date and shift team, view a table of all stops with their causes and durations, see a bar chart showing which causes are most frequent, and view the daily OEE (Overall Equipment Effectiveness) metric that measures productive time versus downtime.

2.2 Criticism of the existing system

2.2.1 Limited Data Analysis

The existing system successfully records stops but provides limited analytical capability. The manager interface shows a bar chart of stop causes and a table of stops for the selected period, but it does not show trends over time. A manager viewing one week of data sees aggregated numbers but cannot easily see whether the situation is improving or deteriorating day by day. The interface does not enable comparison between teams or shifts, making it difficult to identify performance patterns.

2.2.2 No Real-Time Manager Updates

A significant limitation is that the manager web interface does not automatically update. When a manager opens the dashboard to check production status, they must manually click a filter button to retrieve current data. If new stops occur after they view the data, these are not automatically displayed—the manager sees the situation as it was at their last manual refresh. In production environments, delays in visibility can mean slower response to problems.

2.2.3 Limited Performance Metrics

The system primarily reports OEE and stop causes. Modern production monitoring typically tracks additional metrics to provide a complete picture. The system does not show trends in OEE over time, separate availability from performance metrics, or provide any predictive capability to identify potential problems before they occur. A manager cannot easily identify whether equipment health is improving or deteriorating, or whether specific causes show patterns that might indicate underlying issues.

2.2.4 Inflexible User Interface and Limited Access

All managers see the same view regardless of their role or needs. The interface is designed for desktop computers and is not optimized for mobile devices, meaning managers cannot check production status from a smartphone while away from their office. There is also no concept of custom dashboards where different users might want different information.

2.2.5 Data Quality and Verification

The system relies on operators to manually select stop causes from a dropdown menu. There is potential for human error or inconsistency in cause selection. Once a cause is recorded, there is no verification process or audit trail to show who entered the data or when. If a data discrepancy is discovered later, there is no way to trace back what happened.

2.2.6 Lack of Security and Integration

The existing system has no user authentication or access control. Anyone with the IP address can access all production data without logging in. The system also cannot export data to other formats or systems. If a manager needs to perform advanced analysis in Excel or if another department needs production data, it must be manually copied or requested.

2.2.7 Scalability Concerns

The Node-RED platform handles all functions in a monolithic design: reading from the PLC, storing in the database, and serving both the operator tablet and manager web interface. As data volume grows or additional machines are added, this architecture may face performance or reliability issues, since any slowdown in one function could impact others.

3 Modeling languages: SysMl

SysMl, or Systems Modeling Language, is a graphical language used by MBSE (Model-Based Systems Engineering) practitioners to create system models and communicate ideas about system designs to stakeholders. It is a language that has a grammar and vocabulary consisting of graphical notations that have specific meanings. The purpose of SysML is to visualize and communicate a system's design among stakeholders [5].

The grammar and notations of SysML are defined in a standards specification published by the Object Management Group, which is a consortium of computer industry companies, government agencies, and academic institutions. SysMl is an extension of a subset of the Unified Modeling Language (UML), and its complete definition requires referral to parts of the UML specification document as well [5].

We can name a few types of SysMl diagrams:

- **Behavior diagrams [5]**
  - Activity diagram
  - State machine diagram
  - Sequence diagram
  - Use case diagram
  - Communication diagram
- **Structure diagrams [5]**
  - Block definition diagram
  - Internal block diagram
  - Parametric diagram
  - Package diagram
  - Composite structure diagram
- **Requirement diagram[5]**

Conclusion

Overall, chapter 1 has provided the necessary foundation for the subsequent chapters, enabling a comprehensive exploration of the problem and the development of an effective solution. By understanding the host company, the specific problem through the case study, and the modeling language employed, you the reader will be equipped with the context and knowledge required to delve deeper into the next chapters presented in this report.

Chapter 2

SQL Optimisation

Introduction

This chapter documents the complete optimization journey of our MySQL database that collapsed under its own computational weight from the first symptoms of degradation, through systematic diagnosis, to a confirmed final architecture that achieves sub-second response times at full production scale.

1 Context and System Overview

1.1 Context

At the start of this project, the database contained a working implementation. The queries returned correct results. The system appeared functional. However, as the dataset grew during testing toward realistic production volumes, a fundamental problem emerged: what worked at a few thousand rows began to show alarming degradation at tens of thousands, and became completely unusable when approaching the million-row scale that real multi-year industrial deployments would inevitably reach.

1.2 System Overview

1.2.1 Database Structure

The production database consists of four main tables: stops (machine downtime events), causes (reason categories), vitesse (machine speed tracking), and metrage (distance metrics). This optimization focuses on the stops and causes tables, which form the core of the dashboard queries and represent the primary performance challenge.

The stops table stores one record per machine stop event:

Table 2: Core stops table structure

Column	Type	Description	Constraint
`id`	INT	Unique identifier	PK
`Jour`	DATE	Stop date	NN
`Debut`	TIME	Start time	NN
`Fin`	TIME	End time	NN
`cause`	INT	Cause reference	FK

1.2.2 Production Context

The industrial context imposes specific constraints that directly shape every optimization decision:

- **Three production teams (*équipes*):** Team 1 works 06:00–14:00, Team 2 works 14:00–22:00, Team 3 works 22:00–06:00.
- **Shifted production day:** Stops occurring before 06:00 belong to the *previous* calendar day's Team 3 shift. This introduces a derived *production day* concept that differs from the calendar date.
- **Insert-heavy workload:** 2,000–7,000 stop events per day, inserted automatically via machine controllers. Stops are never deleted.
- **Dashboard usage:** Up to three supervisors query date-range filtered views throughout each shift. Sub-second response is the target.

1.2.3 The Key Metric: Percentage of Daily Downtime (PCT)

The primary computed field in every query is the percentage of total downtime that each stop represents within its production day and team:

$$\text{PCT} = \frac{\text{stop duration (seconds)}}{\text{total team downtime for that production day}} \times 100 \tag{2.1}$$

This metric is deceptively simple to state but computationally expensive at scale. The denominator requires aggregating all stops for the *same team on the same production day* for every single row in the result. How that aggregation is implemented is the central performance challenge of this entire part.

2 Query A: The Original Implementation

2.1 Implementation

The original query was written to correctly compute all required fields for the dashboard: stop times, duration, team assignment, cause information, and the PCT metric. The implementation is shown in Listing 2.1.

```sql
SELECT
    s.Debut AS startTime,
    s.Fin AS stopTime,
    CASE
        WHEN s.Fin IS NULL THEN NULL
        WHEN s.Fin >= s.Debut
            THEN TIME_TO_SEC(s.Fin) - TIME_TO_SEC(s.Debut)
        ELSE (TIME_TO_SEC(s.Fin) + 86400) -
            TIME_TO_SEC(s.Debut)
    END AS duree_secondes,
    (SELECT c.name FROM causes c
        WHERE c.id = s.cause_id LIMIT 1) AS cause,
    (SELECT c.affect_trs FROM causes c
        WHERE c.id = s.cause_id LIMIT 1) AS affect_trs,
    ROUND(
        CASE
            WHEN s.Fin IS NULL THEN NULL
            WHEN s.Fin >= s.Debut
                THEN TIME_TO_SEC(s.Fin) - TIME_TO_SEC(s.Debut)
            ELSE (TIME_TO_SEC(s.Fin)+86400)-TIME_TO_SEC(s.Debut)
        END
    ) * 100.0
    / NULLIF((
        SELECT SUM(
            CASE
                WHEN sx.Fin IS NULL THEN 0
                WHEN sx.Fin >= sx.Debut
                    THEN TIME_TO_SEC(sx.Fin) -
                        TIME_TO_SEC(sx.Debut)
                ELSE (TIME_TO_SEC(sx.Fin)+86400)-
                    TIME_TO_SEC(sx.Debut)
            END
        )
        FROM stops sx
        WHERE DATE_FORMAT(
            IF(sx.Debut < '06:00:00',
                DATE_SUB(sx.Jour, INTERVAL 1 DAY), sx.Jour),
            '%Y-%m-%d'
        ) = DATE_FORMAT(
            IF(s.Debut < '06:00:00',
                DATE_SUB(s.Jour, INTERVAL 1 DAY), s.Jour),
            '%Y-%m-%d'
        ), 0), 2
    AS pct
FROM stops s
WHERE DATE_FORMAT(
    IF(s.Debut < '06:00:00',
        DATE_SUB(s.Jour, INTERVAL 1 DAY), s.Jour),
    '%Y-%m-%d'
```

```
) BETWEEN @from AND @to;
```

Listing 2.1: Query A — Original Implementation

2.2 Initial Behavior

At small row counts, Query A functions correctly and returns accurate results. For a development environment or early testing phase, it raises no alarms.

But as the data grows, the pattern in Table 2 reveals something alarming: each time the row count doubles, the execution time does not double — it quadruples. This is the signature of $O(n^2)$ complexity.

Table 3: Query A measurements

Rows	Time (s)	Observation
100	0.291	Fast, no concern
500	7.812	Slightly slow but tolerable
1,000	28.347	First warning sign
1,500	67.093	Clearly problematic
2,000	114.156	Unusable in production
3,000	273.987	
5,000	701.245	
10,000	3,041.678	
14,000	5,512.934	Testing stopped here
1,000,000	$\approx$ 329 days	Projected via $k \cdot n^2$

2.3 The Root Cause: A Fully Correlated PCT Sub-query

The measured growth follows a quadratic pattern:

$$T_{\text{Query B}}(n) \approx k \cdot n^2 \tag{2.2}$$

where n is the number of rows and k is a constant representing the per-comparison cost. Compounding this, the correlated subquery uses a non-sargable predicate — `DATE_FORMAT(IF(...))` wraps the column in a function call, making it impossible for MySQL to use any index even if one existed on `Debut` or `Jour`.

The fundamental problem with Query B is the PCT denominator computation. For every single row s in the outer query, MySQL fires a full scan of the entire `stops` table to compute the daily total.

2.4 SHOW PROFILE Analysis

To obtain quantitative evidence of where time was being spent, MySQL's SHOW PROFILE was used with a 5-row LIMIT on the outer query. Even with only 5 outer rows processed, the profile captured 93 execution cycles consuming 1,759 ms in total.

Table 4: SHOW PROFILE results for Query B .

Phase	**Duration (ms)**	**% of total**
executing (correlated subquery loop)	1,759.6	99.94%
freeing items	0.357	0.02%
logging slow query	0.489	0.03%
closing tables	0.033	0.00%
cleaning up	0.065	0.00%
Other	0.101	0.01%
Total	**1,760.6**	**100%**

This confirms that almost all time (99.94%) is spent inside the correlated subquery loop, with all other database operations (table opening, memory allocation, logging, cleanup)account for just 1.045ms combined, validating the O(n2) diagnosis and highlighting the denominator computation as the dominant bottleneck.

Netography

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